Software Requirements Specification (IEEE 830-style)

Product: Agentforce Enterprise Bridge **Version:** 1.0 **Status:** Approved for MVP

1. Introduction

Purpose: specify a governed integration that lets Agentforce-style agents act on enterprise telemetry safely. **Scope:** ingestion, grounding, planning, governed action, audit. **Out of scope:** UI console, production Kafka, Data Cloud licensing.

2. How requirements were gathered (illustrative engagement)

Personas below model a typical discovery; replace with real names/outcomes when presenting a real engagement.

Stakeholder	Ask (verbatim style)	Classification	Resolution
Business sponsor (COO)	"AI that resolves disputes automatically."	**Ambiguity** — what is "resolve", which decisions?	Decision-classification workshop → autonomy tiers (FR-5, FR-6)
Compliance lead	"Nothing financial without a human; prove it."	Clear constraint	HITL gate + audit (FR-6, FR-8)
CRM admin	"Don't break our org; no new licences."	Constraint	Free-tier objects, permission set, SFDX (FR-9, C-2)
Customer architect	"Will it survive upstream schema changes and 429s?"	**Uncertainty** — measurable by spike	Contract + DLQ, breaker/backoff spike (FR-2, NFR-2)
Security	"No PII to models or CRM; least privilege."	Clear constraint	PII redaction, RBAC (FR-4, FR-7)

Ambiguity = the requirement is unclear and is resolved by asking people. *Uncertainty* = the answer is unknown and is resolved by experiments/measurement.

3. Functional requirements

ID	Requirement	Verified by
FR-1	Accept telemetry via `POST /v1/events`; sync (`wait=true`) and async modes	`test_api_end_to_end`
FR-2	Reject payloads violating the versioned schema; retain in DLQ	`test_bus_flow`
FR-3	Idempotent ingest by `event_id`; 429 when saturated	`test_bus_flow`, `test_backpressure_429`
FR-4	Redact SSN, card (Luhn), email, phone before prompts/audit/CRM	`test_redact*`, `test_pii_redacted_before_audit`
FR-5	Resolve account name variants to one entity; scope context per account	`test_grounding_*`
FR-6	Risk-tiered actions; HIGH requires approval by a different human, with TTL	`test_hitl_*`, `test_approval_store_ttl`
FR-7	RBAC: VIEWER/AGENT/APPROVER/ADMIN; fail-closed auth	`test_rbac`, `test_api_end_to_end`
FR-8	Audit every decision locally and in `Bridge_Audit_Log__c`; retry on outage	`test_audit_outbox`
FR-9	Create/escalate Cases via Salesforce REST using OAuth 2.0	`test_salesforce.py`
FR-10	Expose action catalog/OpenAPI for Agentforce External Services	`test_registry`
FR-11	LLM planner optional; degrade to rules on any failure	`test_resilient_planner`

4. Non-functional requirements

ID	Requirement	Verification
NFR-1	Offline p95 pipeline latency < 5 ms; ≥1k events/s/core	`bridge bench`
NFR-2	Survive 429/5xx/network/401 with bounded retries; breaker prevents cascade	`test_retries_*`, `test_circuit_opens`

ID	Requirement	Verification
NFR-3	Memory bounded (queue, dedupe, history, DLQ, audit, approvals)	design + unit tests
NFR-4	No secrets in repo; masked in logs/repr	`test_settings_env`, `.env.example`
NFR-5	100% line+branch coverage; Google style; CI matrix 3.11–3.13	CI
NFR-6	Health/readiness/metrics endpoints	`test_api_*`

5. Constraints & assumptions

C-1 Python ≥3.11. C-2 Free Developer Org; client-credentials flow needs My Domain URL. C-3 Single replica for MVP (in-memory approvals). A-1 Upstream sends UTC timestamps.

6. Acceptance criteria

Demo shows: PII stripped, “ACME Corp.” merged with “Acme Corporation”, low-risk Case auto-created, $25k review held until a different human approves, audit rows for all, CI green at 100%.